

What Do Defaults Tell Us About Discount Rates? Revealed Preference from 20 Years of Credit-Bureau Data

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Job Talk

2026–27 Market

Roadmap

- 1. The question and the data**
2. A sufficient statistic for the discount rate
3. Baseline estimates — and every reason not to believe them, addressed
4. Heterogeneity: who discounts, where, and when
5. Aggregation and what it means

The discount rate is the primitive of household finance

- ▶ Every intertemporal decision a household makes — saving, borrowing, defaulting, refinancing, retiring — is priced off one parameter
- ▶ Yet we mostly measure it by **asking hypothetical questions** (elicited rates: 30–100%+, unstable) or by structural estimation inside heavily parameterized models
- ▶ This paper: measure it from **what borrowers actually do at the highest-stakes margin they face** — default

Default is an intertemporal trade

- ▶ A borrower who defaults **keeps the remaining payments** (money today and tomorrow) and **accepts the consequences** (chargeoff, collections, credit exclusion — money later)
- ▶ Where the borrower stops paying therefore *reveals* the rate at which they trade the future for the present
- ▶ The paper turns this observation into a sufficient statistic that needs no structural model of the household

Preview of answers

- ▶ Revealed-preference annual discount rate: $\approx 27\%$ at the three-year horizon ($\hat{\beta} = 0.787$, s.e. 0.003) — and the marginal-utility correction is *measured* to be nil
- ▶ An order of magnitude above risk-free rates; far below elicited rates; consistent with behavior in the credit-card literature
- ▶ The heterogeneity is the economics: rates decline monotonically in credit score (26.7% $\rightarrow$ 23.2%) and in income (25.8% $\rightarrow$ 21.8%); geography spreads β across [0.73, 0.84]
- ▶ And the headline twist: different loan horizons trace out a **steeply declining discount curve** — within person — a discount rate schedule

The data: the best you could ask for on this question

- ▶ **Equifax ACRO archive, 10% sample of US credit files:** every trade line, **monthly snapshots 2005–2024** (240 archives)
- ▶ Auto loans: **6.2 million** fixed-term loans, **277,026 defaults** usable for estimation, all 48 continental states + DC
- ▶ Same archive, same estimator, three more contract families: **58M first mortgages, 4.9M home-equity, 72M personal installment** — horizons from under two years to seventeen
- ▶ Terms, payments, balances, chargeoffs, timing — everything the estimator needs, at population scale, for 20 years

Contribution relative to three literatures

- ▶ **Measured discount rates** (elicitation, field experiments): we replace stated choices with consequential ones, at 6M-loan scale
- ▶ **Consumer default** (strategic default, liquidity default): we use default *timing* as information about preferences, not just a cost to be explained
- ▶ **Household finance measurement** (payment targeting, bureau-data methods): we contribute a practitioner's toolkit — rate imputation from quantity fields, power-preserving sample design — reusable beyond this paper

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Two periods, one identity

- ▶ Defaulting borrower keeps remaining payments A , loses chargeoff consequences B , with continuation default risk p_2
- ▶ Indifference at the observed stopping point delivers the **exact identity**

$$\frac{A}{(1 - p_2) B + A} = \beta \lambda$$

- ▶ β : the discount factor. λ : relative marginal utility of the default state — the correction term the literature usually assumes away

From identity to estimator

- Implementation: two fixed-effects logits (default on log payment; default on log total obligation) within score-ventile $\times$ zip-by-month cells
- $\hat{R} = \hat{\gamma}_L / (\hat{\gamma}_L + \hat{\gamma}_m)$, annualized through mean defaulter horizon $\widehat{\Delta T}$:

$$\hat{\beta} = \hat{R}^{1/\widehat{\Delta T}}$$

- Identification: cross-sectional variation in payment size against obligation size, within cells that absorb local shocks and creditworthiness

The theory extends as far as anyone will push it

- ▶ T -period and continuous-time generalizations: the identity survives as the envelope condition of the optimal stopping problem (Appendix; available for the whiteboard)
- ▶ Heterogeneous agents: aggregation theory for what a population β distribution implies for measured aggregates — pooled estimation recovers a sensitivity-weighted average of type-level statistics (exact aggregation proposition)
- ▶ Marginal-utility correction: **measured directly in the panel** — the key input is the persistence of the marginal borrower's state, and 21.4M delinquency transitions put it at $\rho = 0.992 \Rightarrow \lambda = 1.00$, correction nil (the literature's lower-persistence calibrations move the rate to no less than 25%)

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Baseline estimate

- ▶ $\hat{\beta} = 0.787$ (s.e. 0.003) $\Rightarrow \approx 27\%$ **per year** at the three-year horizon
- ▶ The one approximation is measured away: at the panel's own distress persistence, $\lambda = 1.00$ — the correction is nil
- ▶ And the clock is not doing the work: the sensitivity ratio $R = 0.465$ would need $T = 18.8$ years on five-year loans to be consistent with $\beta = 0.96$
- ▶ Benchmarks: risk-free $\sim 2\text{--}4\%$; credit-card APRs $\sim 20\text{--}25\%$; elicited experimental rates $30\text{--}100\%+$

▶ measured correction ▶ is it just T ? ▶ precision

Power here is a theorem

$$\text{se}(\hat{\beta}) = \frac{c}{\sqrt{n_{\text{defaults}}}}$$

- ▶ Rare-events information: every entry of the estimator's covariance scales with $1/n_d$ — **loans are free, defaults are power**
- ▶ Validated on 136 geographic clusters: log-log slope -0.615 (0.077); conditional on defaults, the loan count contributes **exactly nothing**: -0.016 (0.153)
- ▶ Consequence: every sample split in this paper is designed at the default margin (equal-default bands, default-floor clusters)

Threat 1: “you never observe the interest rate”

- ▶ Correct — no credit bureau records prices; the rate field is a payment-status code
- ▶ Answer: the contract's quantity side (L, m, T) **overidentifies** the rate — annuity inversion at origination, balance-path recovery from monthly snapshots
- ▶ Both validated on simulated bureau data with known rates; the two methods fail on **disjoint** reporting quirks, so their agreement is a data-quality diagnostic (Appendix: practitioner's guide)

▶ [error taxonomy](#)

Threat 2: “payment size is endogenous”

- ▶ We killed our own first instrument, in public: rate levels at origination fail three redesigns (vintage confounding — a purely cohort-level instrument cannot be rescued)
- ▶ The living design: **lender-menu leniency** — the staggered 60→72→84-month term expansion and lender size menus give two leave-one-out instruments in the *same* sample as the FE strategy; first stages $F = 221$ and $F = 47,000$
- ▶ Win condition: IV and FE agree quantitatively where both run, apples to apples — then FE carries the heterogeneity [**PLACEHOLDER: lender-FE iteration — final table in flight**]

▶ [the instrument graveyard](#)

Threat 3: “defaulters are selected”

- ▶ Of course — the estimator *conditions* on the default margin; the question is what changes across comparisons
- ▶ Cross-group comparisons hold the margin fixed by design (equal-default bands); geographic correlations are re-estimated under default-composition controls
- ▶ Example below: the income–patience correlation **dies** under the selection control; the poverty and partisanship correlations **survive** — the paper reports both honestly

▶ [full table](#)

Threat 4: “they just can’t pay”

- ▶ **65% of 2.46M auto defaulters keep paying other creditors** six months *after* the chargeoff (10.7 other trades on average; only 1.6% default on everything) — the modal default is a portfolio choice
- ▶ $\hat{\beta}$ is **flat across local labor shocks**: 0.774 / 0.786 / 0.789 by unemployment-change tercile
- ▶ The same borrower reveals 29% on personal loans, 12% on autos, 1% on the mortgage — **one budget cannot bind selectively**
- ▶ Distress is measured near-permanent ($\rho = 0.992$) — the transitory-shock story contradicts the data; the permanent-state story *is* the measured λ

▶ Monte Carlo: why $\hat{\beta}$ can’t acquit itself

Threat 5: “they just expect to be rich”

“Everyone was a temporarily embarrassed capitalist.”

— John Steinbeck, “A Primer on the Thirties,” *Esquire*, 1960

(the popular “temporarily embarrassed millionaires” is Ronald Wright’s 2004 paraphrase)

- ▶ The objection: high borrowing rates reveal expected income growth, not impatience
- ▶ Three answers, in increasing strength: distress at the default margin is measured near-permanent ($\rho = 0.99$, 0.26% of spells cure per quarter); the growth calibrations large enough to matter are refuted by the data’s own bound ($\lambda \geq 0.47$); and the SCF lower bounds hand every household its *most optimistic* lifecycle income path — the 16–20% floor survives
- ▶ Even a temporarily embarrassed capitalist should not pay 25% APR once the model has priced in the capital

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The economics lives in the cross-section of β

- ▶ A single aggregate discount rate answers no policy question — **who** is impatient governs the incidence of every intertemporal policy: moratoria, minimum payments, forbearance, retirement defaults
- ▶ This section: credit score, income, loan type, geography, political geography, and time
- ▶ Power-preserving design: every cell flagged by its default count, nothing silently pooled

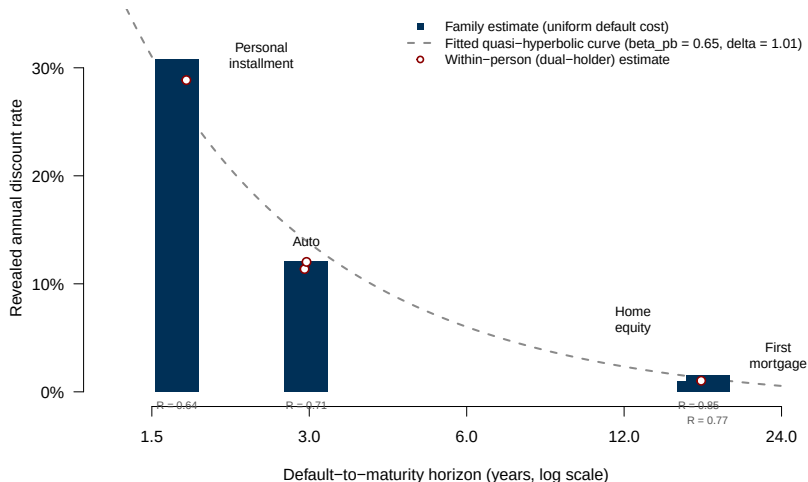
By credit score: monotone, precise, and modest

- ▶ Varying-coefficient design: the two logits carry heterogeneity in their derivatives, so every default informs every cell — standard errors of **0.4pp** where cell-splitting gives multiples
- ▶ Three-year rates decline monotonically across score deciles: **26.7% (d1) → 23.2% (d10)**
- ▶ The gradient is real but *modest* — high default rates among low-score borrowers are mostly not impatience

By income: the patient rich, measured

- ▶ Same design, income quartiles (at median score): **25.8% → 21.8%** from the bottom to the top quartile
- ▶ Individual-level gradient, holding score and region — distinct from (and cleaner than) the zip-level income correlate, which dies under the selection control
- ▶ Correction invariance: a common λ cannot generate cross-group gradients

The revealed discount curve, traced loan by loan



Four families, one estimator, uniform default cost — the annual rate falls from **31% at a 1.7-year horizon to 1% at seventeen years**, and the dual-holder markers show it **within person**. The measurement is a schedule.

Geography: 136 power-designed clusters, 48 states

- ▶ Sub-state clusters built on zip3 adjacency to a default floor; estimated cluster-by-cluster with the full estimator
- ▶ Cross-cluster spread is **large**: β p10–p90 [0.733, 0.835] — annual rates from $\sim 20\%$ to $\sim 36\%$
- ▶ Decomposition: 84% of the cross-cluster variance in $\log \beta$ comes from the default sensitivities alone; the horizon contributes 2%

What geography correlates with

- ▶ Merged with IRS SOI zip3 covariates (loan-count weighted):
- ▶ **Poverty (EITC share):** $\rho_w = -0.37$, slope -0.42 (0.09) — poorer areas show *lower* measured discount rates; survives default-composition control
- ▶ Income level: raw positive, **dies under the selection control** — reported as a selection artifact
- ▶ Unemployment share, itemizer share: positive, state-FE fragile

Political geography: patience leans Republican

- ▶ County presidential returns (2008–2024) crosswalked to clusters
- ▶ 2020 GOP two-party share: $\rho_w = -0.19$ (Spearman -0.30); slope -0.088 (0.040), **strengthens** under the default-composition control (-0.101 (0.038))
- ▶ The 2016–2020 **swing is a precise zero** ($\rho_w = 0.01$) — the partisan *level* correlates with patience; the Trump-era movement is orthogonal to it
- ▶ Five-cycle panel (2008–2024) in estimation **[PLACEHOLDER: geography $\times$ cycle betas — within-cluster changes, lead-lag structure]**
- ▶ Companion analysis: enters the manuscript with the next estimation cycle; replication outputs in the project repository

Across time: patience is rising

- ▶ State $\times$ cycle cells, five presidential cycles: median β climbs **0.719** $\rightarrow$ **0.764** $\rightarrow$ **0.779** $\rightarrow$ **0.774** $\rightarrow$ **0.808** from 2005–08 to 2021–24
- ▶ Rates at the default margin fell from $\sim 39\%$ to $\sim 24\%$ over two decades — composition, credit access, or preference drift: the panel can ask
- ▶ Recession cohorts run more impatient (0.731 vs. 0.792) — a composition effect the paper flags and does not over-read

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From micro β to macro objects

- ▶ Heterogeneous-agents aggregation: measured aggregates load on the *patient tail*, so population medians and macro calibrations answer different questions
- ▶ The 27% median coexists with standard macro calibrations, and the aggregation proposition states exactly how

What it means

- ▶ Incidence: debt relief and forbearance are transfers *to* the impatient margin, and the margin is now measured
- ▶ Pricing: lenders setting minimum payments face a distribution of β this paper measures directly
- ▶ Measurement: the toolkit (rate imputation, power-preserving splits, the exact identity) transfers to any bureau-data setting

Conclusion

- ▶ Defaults reveal discount rates: $\approx 27\%/year$ at the three-year horizon — and a **steeply declining schedule** across horizons, within person
- ▶ The best available data, attacked with an exact identity, a *measured* correction, designed power, and every objection diagnosed in the open
- ▶ The discount curve (four bars, one dashed line) is the picture to leave the room with ▶ [the curve](#)

Backup arsenal

Organized for the fight: theory, calibration, power, imputation, selection, data plumbing.

Backup: exact identity, full derivation

- ▶ Two periods: implicit differentiation of the period-1 threshold gives $\partial p_1 / \partial L = f(y_1^*) \beta (1 - p_2) \mathbb{E}[u'(C_2^P) \mid y_2 > y_2^*] / \Delta u'_1$ and $\partial p_1 / \partial m$ as the current-payment analogue
- ▶ The ratio cancels the margin density and the utility scale, leaving $\beta \lambda$
- ▶ T periods: the same identity survives as the envelope condition of optimal stopping, provided the punishment has a flow component and good standing carries the option to default later

Backup: continuous-time statement

- ▶ Continuous time: default is a stopping time on the income process, and the identity appears as the smooth-pasting condition at the default boundary
- ▶ The response of stopping to a payment perturbation at horizon s is $D(s)$ times the survival-weighted marginal utility at s — ratios across horizons identify the discount function nonparametrically
- ▶ Two horizons point-identify quasi-hyperbolic (β_{pb}, δ) ; three make the exponential null testable

Backup: the marginal-utility correction, measured

- ▶ The key input is the persistence of the marginal borrower's state: 21,350,801 delinquency-spell transitions, **0.26% cure per quarter** $\Rightarrow \rho = 0.992$ annualized
- ▶ At measured ρ : $\lambda = 1.00$ at the Blundell–Pistaferri–Preston pass-through — **correction nil**; the headline stands as measured
- ▶ Literature-persistence sensitivity ($\rho = 0.95$, STY 2004): $\lambda = 0.95$, corrected rate 25% — the floor across every admissible calibration
- ▶ Internal-consistency bound: identification requires $\lambda \geq \hat{\beta}^T = 0.47$, refuting exactly the calibrations that would generate large corrections

[◀ back: baseline](#)

Backup: the clock contributes little

- ▶ Decompose cross-group variance of $\log \beta$ into sensitivity (R) and horizon (T) contributions: geography **84% R -only** (rank correlation 0.97 at fixed T); state-cycle 89%
- ▶ Levels are T -proof: $R = 0.465$ requires **$T = 18.8$ years on five-year loans** to be consistent with $\beta = 0.96$; a 20% error in T moves the headline 27% $\rightarrow$ 22%
- ▶ Where T *does* carry magnitude — across loan families — that is the term-structure result, promoted to a finding

[◀ back: baseline](#) [◀ back: the curve](#)

Backup: the estimator under liquidity-driven default

- ▶ Ground rule: in a simulated pure-liquidity economy, $\hat{\beta}$ lands at **0.78, not 0** — obligations load mechanically on payments, so $\hat{\beta}$ alone cannot referee
- ▶ Mixture path (true $\beta = 0.95$): $\hat{\beta}$ falls $0.887 \rightarrow 0.783$ as the liquidity share goes $0 \rightarrow 1$
- ▶ Default-rate-matched surface, true $\beta \in [0.55, 0.99] \times$ shares up to 0.50: **no cell reaches the measured 0.787** — patient preferences plus admissible contamination cannot manufacture the estimate

[◀ back: threat 4](#) [◀ back: the curve](#)

Backup: two horizons identify (β_{pb}, δ)

- ▶ Quasi-hyperbolic $D(t) = \beta_{pb} \delta^t$: with horizon factors $b(T) = D(T)^{1/T}$,
$$\ln \beta_{pb} = \frac{\ln b(T_1) - \ln b(T_2)}{1/T_1 - 1/T_2}$$
- ▶ Dual holders: auto $\times$ mortgage $\Rightarrow \beta_{pb} = 0.71, \delta = 1.01$; auto $\times$ personal $\Rightarrow \beta_{pb} = 0.55, \delta = 1.09$ — **large present bias, long-run rate ≈ 0** within measurement error
- ▶ Estimand discipline: every cross-person object is *horizon-labeled*; cross-horizon statements are statements about the curve; the schedule-proof heterogeneity object is the joint distribution of (β_{pb}, δ)

[◀ back: the curve](#)

Backup: power law — matrix statement

$$\text{Avar}(\hat{\beta}) = \nabla g' \Sigma \nabla g, \quad \Sigma = V/n_d, \quad \mathcal{I}(\gamma) \approx \mathbb{E}[n_d] \cdot M$$

- ▶ Rare-events logit information proportional to expected events (King–Zeng 2001); M = default-weighted design moment
- ▶ Horse race across 136 clusters: $\ln n_d$ coefficient -0.608 (0.102); $\ln n$ coefficient -0.016 (0.153)

[◀ back: baseline](#)

Backup: rate imputation — error taxonomy

- ▶ Method 1 exact except: odd first period (+18bp), balloons (adjusted inversion restores exactness)
- ▶ Method 2 exact except intermittent stale balances (−0.1bp median rule) and curtailments (−0.6bp median; mean version explodes)
- ▶ Financed fees: no bias; a ~77bp note-rate/APR wedge at 2% fees — imputed rates lower-bound Reg-Z APRs

[◀ back: threat 1](#)

Backup: selection diagnostics, full table

- ▶ Poverty (EITC share): slope -0.42 (0.09) without the default-composition control, and the correlation survives with it
- ▶ Zip-level income: positive in the raw cross-section and gone under the control — reported as a selection artifact
- ▶ Partisanship level: -0.088 (0.040) raw, -0.101 (0.038) controlled; the 2016–2020 swing is a precise zero

[◀ back: threat 3](#)

Backup: the instrument graveyard, and the living design

- ▶ GS3-at-origination, three repairs, three failures: vintage controls ($\beta = 1.14$, CF = 4.7), AR(1) innovations (first-stage $F = 0.0$), imputed-rate target ($\beta = 1.07$) — a purely cohort-level instrument cannot be rescued
- ▶ Menu leniency: first stages $F_{dur} = 221$, $F_{size} = 47,288$ (lender-clustered); naive CF flips γ_m — long-term-menu lenders are subprime, menus proxy clientele
- ▶ The pre-registered fix: lender FE, identification from *menu changes within lender* [PLACEHOLDER: final agreement table]

◀ back: threat 2

Backup: data plumbing that survives review

- ▶ Single deterministic pipeline, fail-fast probes, restartable extracts; every exhibit regenerated from raw archives
- ▶ Known traps handled: zip zero-padding, integer overflow in population-weighted merges, CT's 2024 FIPS redistricting, left-truncated pre-2005 cohorts, censored 2021+ cohorts

Backup: what the archive contains

- ▶ 73 fields per trade-month; flags for every loan family; balloon, deferral, and narrative codes (fixed vs. variable rate)
- ▶ 240 monthly archives $\times$ 10% of US credit files, 2005–2024